

Review

Leveraging Artificial Intelligence in Project Management: A Systematic Review of Applications, Challenges, and Future Directions

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Abstract: This article presents a systematic literature review exploring the integration of Artificial Intelligence (AI) methodologies in project management (PM). Key applications include cost estimation, duration forecasting, and risk assessment, which are critical factors for project success. This review synthesizes findings from 97 peer-reviewed studies published between 2011 and 2024, using the PRISMA methodology to ensure rigor and transparency. AI techniques such as machine learning, deep learning, and hybrid models have exhibited their potential to enhance PM techniques across projects' phases, including planning, execution, and monitoring. Decision trees are created to represent the application of AI methodologies in various PM stages and tasks to facilitate understanding and real-world implementation. Among these are hybrid AI models that enhance risk assessment, duration forecasting, and cost estimation, as well as categorization based on project phases to optimize AI integration. Despite these advancements, there are still gaps in addressing dynamic project environments, validating AI models with real-world data, and expanding research into underexplored phases like project closure.

Keywords: project management (PM); Artificial Intelligence (AI); cost estimation; duration forecasting; risk assessment; machine learning (ML); hybrid models; decision trees; PRISMA methodology; systematic literature review



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1. Introduction

Project management (PM), according to PMBOK GUIDE 7th edition [1], is defined as the application of knowledge, skills, tools, and techniques to manage and lead activities. PM methodology describes project success as the attainment of project objectives and the completion and acceptance of all project deliverables by the client [2]. In addition, Turner and Xue [3] argue that a project can be successful when it offers meaningful benefits, aligns with the agreed deliverables, and follows the specified schedule and budget. The knowledge of suitable information and technology is essential for achieving success in projects [4]. Due to technological improvements, the field of PM has seen rapid evolution in recent years. Artificial Intelligence (AI) is one of the most revolutionary technologies making a significant impact in this industry. The study of intelligent computational agents, or “computational intelligence”, is a component of AI [5,6]. AI has demonstrated the ability to direct projects, automate PM duties, and support decision-making processes [7,8]. The applications of AI in projects are diverse, serving to reduce risks, monitor their progress, and identify abnormalities and exceptions in them [7,9]. Costing, schedule control, activity management, and resource planning can be supported by leveraging AI in PM [7,10]. Martínez and Fernadez-Rodriguez [11], through their research, indicated that AI tools

exhibit greater accuracy compared to traditional methods, yet they remain supplementary to traditional tools. Project managers can greatly benefit from the integration of AI tools in overseeing and managing their projects.

This research offers a systematic literature review (SLR) to thoroughly examine the role of AI in PM. By analyzing the utilization of AI methodologies in critical project success factors (CSFs), such as cost estimation, duration forecasting, and risk assessment, this study focuses on enlightening researchers, stakeholders, and practitioners about the potentials of AI in advancing PM techniques. Cost estimation, duration forecasting, and risk assessment are often considered CSFs in PM. These elements are vital for the successful delivery of a project across its lifecycle. According to the research by Pinto and Slevin [12], these factors are consistently highlighted as crucial during the planning and execution phases, as they directly influence a project's ability to meet its objectives within the defined constraints of time and cost [13].

The present article delves deeply into the literature on the application of AI in PM, encompassing various approaches, tools, and challenges. It aims to analyze the applications of AI in PM, identify gaps and weaknesses in the existing literature, and consolidate knowledge through the methodical collection and evaluation of research and studies. Additionally, it seeks to highlight the advantages of AI—such as increased productivity, decreased risk, and optimized processes—while examining the limitations and challenges organizations must overcome. The findings will assist professionals and researchers in making informed decisions, developing a theoretical framework for integrating AI, and encouraging innovation and sustainable growth in PM.

In the rapidly growing field of Artificial Intelligence (AI) in project management, various studies have explored distinct aspects of this intersection. Notable contributions include the SLRs conducted by Nenni et al. [14] and Prasetyo et al. [15]. These reviews, along with studies by additional researchers such as Costa et al. [7] and Taboada et al. [16], provide a foundation for examining the adoption, integration, and impact of AI technologies in project management processes. Nenni et al. applied an SLR with a bibliometric approach, analyzing 215 studies using PRISMA and VOSviewer to identify trends in AI-driven project management. This study is more focused on mapping AI applications across different Project Risk Management processes and evaluating AI categories and tools commonly used in project risk mitigation. Prasetyo et al. focused on AI adoption in open innovation project management by employing an SLR that emphasized AI's role in enhancing collaboration, decision-making, and knowledge sharing. Taboada et al. examined AI's impact on Industry 5.0 project management, concentrating on AI's application in planning, measurement, and uncertainty performance domains. Their analysis suggested that AI, especially machine learning, has been notably effective in construction and IT projects for forecasting and decision-making. The research of Costa et al. (2022) followed a technology-driven approach, categorizing AI techniques (e.g., machine learning, deep learning, NLP, fuzzy logic) and their specific project management applications. This study focuses more on technical AI methodologies rather than their practical implementation within PM frameworks.

This study is an SLR using the PRISMA framework, ensuring transparency and replicability. It integrates decision tree methodologies to classify AI applications across different project management knowledge areas, including cost estimation, duration forecasting, and risk assessment. By framing the analysis within the PMBOK framework, this research ensures a transparent and replicable process for identifying, screening, and synthesizing the relevant literature and provides a novel roadmap for future AI development, addressing both technical and qualitative challenges, thereby extending the focus beyond quantitative efficiencies emphasized in earlier studies. Moreover, by integrating decision tree methodologies to classify AI applications across PM knowledge areas, this article aims to provide a

visual and analytical framework to understand the distribution of AI applications, highlight gaps, and identify opportunities for future research. These methodological advancements distinguish this study by offering a comprehensive and integrative perspective that bridges theoretical and practical insights.

The structure of this article is as follows: It begins by outlining the methodology used in this research, detailing the steps for collecting and evaluating articles, and explaining how the results are presented through diagrams. Next, the paper provides a comprehensive literature review, summarizing existing studies and their findings on the integration of AI in PM, categorized by the areas of knowledge in PM. In order to effectively present the results of the literature review, decision trees are developed to classify and visually organize the above-mentioned theories, models, and findings. Following this, the article identifies research gaps in the current literature, highlighting areas that require further study. Finally, it concludes with a review of the findings and offers recommendations for future research directions.

2. Materials and Methods

This study conducts a comprehensive review of the global literature on the utilization of AI in PM. Before the start of the search, a review protocol was entered into INPLASY (INPLASY202510041). An SLR, as defined by Tranfield et al. [17] and Denicol et al. [18], is a transparent, rigorous, and detailed methodology that facilitates decision-making. This approach is pivotal for theory-building through the synthesis of knowledge derived from the analysis of studies and methodologies, thereby enhancing the coherence of results and conclusions. It employs explicit and systematic procedures to minimize bias in searching, identifying, evaluating, synthesizing, analyzing, and summarizing the literature. When executed meticulously, this study has the potential to yield reliable findings and robust conclusions, guiding decision-makers and scientific practitioners in making informed decisions and taking appropriate actions [19,20].

The present SLR adopts the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology [7,21]. The PRISMA methodology encompasses a series of steps aimed at achieving desired outcomes, with an information flow similar to that depicted in Figure 1.

The reason behind adopting the PRISMA system lies in its recognized comprehensiveness, its global, widespread use across various fields beyond medical domains, and its ability to enhance consistency among assessments [22], aiding in the conduct of a rigorous systematic review [7].

For data collection, it was decided to utilize two platforms, Web of Science and Scopus, in order to minimize duplicate results. The search terms used on each platform were combinations of “Artificial Intelligence” and “Project Management”; “Machine Learning” and “Project Management”; “Deep Learning” and “Project Management”; “Neural Networks” and “Project Management”; “Natural Language Processing” and “Project Management”; “Fuzzy Logic” and “Project Management”; “Heuristics” and “Project Management”; “Forecasting” and “Project Management”; and finally “Data Analytics” and “Project Management”. Quotation marks were used in all searches to ensure exact matching with these terms. From these searches, a database containing 10,043 records was initially obtained.

PRISMA 2020 flow diagram for new systematic reviews which include searches of databases and registers only

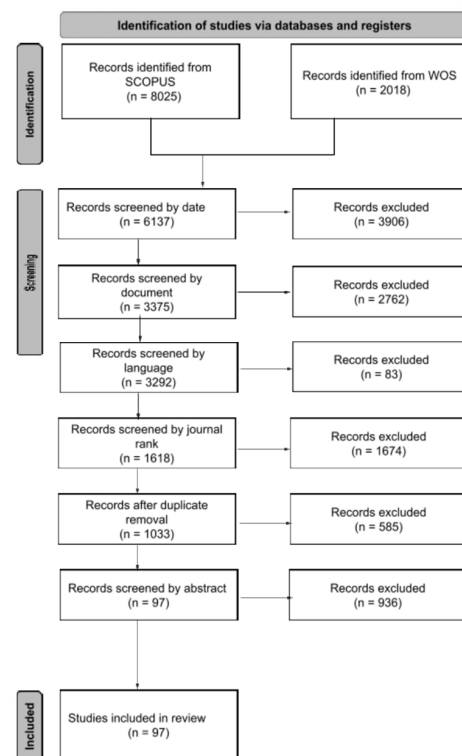


Figure 1. AI in PM—modified PRISMA Flow Diagram [22]. The figure illustrates the systematic review process following the PRISMA 2020 guidelines. Studies were identified from multiple databases and underwent a series of screening steps, including filtering by publication date, document type, language, journal quality, duplicate removal, and abstract relevance. Each step narrowed down the pool of studies to ensure that only the most relevant and high-quality research was included in the final review.

To optimize the dataset to include articles relevant to the research subject of this study, the PRISMA methodology was continued. The first criterion was the publication year of the findings. It was decided to include only findings from 2011 to February 2024 due to the transformation trend in PM resulting from rapid technological advancements in the Fourth Industrial Revolution [23]. This restriction reduced the database by 3906 records, leaving 6137 data points. Next, priority was given to the type of data, retaining only those that were published articles in scientific journals. As a result, 3375 data points were retained (2762 were excluded). Subsequently, for further analysis of abstracts, articles not written in English were excluded (83), resulting in the database containing 3292 data points. Significance was attributed to journal credibility. The criteria chosen were Scimago rankings meeting the conditions for the scientific prestige of journals from Q1 to Q4. To achieve this selection, all journals from the 3292 articles were researched on Scimago and classified, eliminating unrecognized or “non-assigned” journals. Consequently, the remaining articles were published in Q1-category journals, totaling 1618 (1674 were excluded). Duplicate articles that remained were then removed, resulting in a final dataset of 1033 data points (585 were excluded). The remaining articles were read and evaluated based on each article’s title, abstract, and full content, and the process was completed with a final database of 97 entries.

Figure 2, along with Tables 1 and 2, collects the bibliometric results of the selected papers. Figure 2 can be analyzed by examining the points along each axis, which stand for data values for particular years. The scale of the values is shown by the distance from the chart’s center; greater values are represented by larger distances. By comparing changes

over the years shown in the chart, the connecting lines aid in the visualization of trends over time. For example, since 2020, more articles have focused on the involvement of AI in PM. As depicted in Table 1, the natures of the journals of the identified works are diverse: they cover construction and engineering (*Journal of Construction Engineering and Management* is the most cited, followed by *Automation in Construction*), management (*International Journal of PM* is the second most cited in that discipline), and computer science (*Neural Computing and Applications*, *Expert Systems with Applications*, and *Scientific reports*). Table 2 shows that engineering is the main field of the selected papers, with construction industry, computer science, and business and economics following.

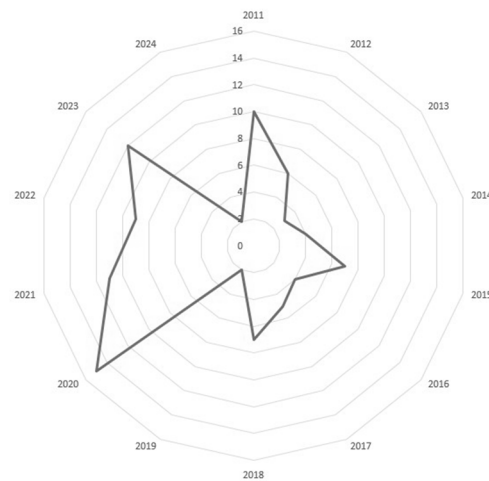


Figure 2. Bibliometric results. Publications per year.

Table 1. Bibliometric results. Articles per journal.

JOURNAL	VALUES
JOURNAL OF CONSTRUCTION ENGINEERING AND MANAGEMENT	13
INTERNATIONAL JOURNAL OF PROJECT MANAGEMENT SUSTAINABILITY (SWITZERLAND)	10
AUTOMATION IN CONSTRUCTION	6
JOURNAL OF MANAGEMENT IN ENGINEERING	5
IEEE TRANSACTIONS ON ENGINEERING MANAGEMENT	4
DECISION SUPPORT SYSTEMS	4
SCIENTIFIC REPORTS	3
IEEE ACCESS	3
AUTOMATION IN CONSTRUCTION	3

Table 2. Bibliometric results. Articles per industry.

INDUSTRY	VALUES
ENGINEERING	53
CONSTRUCTION AND BUILDING TECHNOLOGY	27
COMPUTER SCIENCE	25
BUSINESS AND ECONOMICS	24
OPERATIONS RESEARCH AND MANAGEMENT SCIENCE	14
SCIENCE AND TECHNOLOGY	10
ENVIRONMENTAL SCIENCES AND ECOLOGY	6
TELECOMMUNICATIONS	4
MANAGEMENT	2
ARCHITECTURE CONSTRUCTION AND BUILDING TECHNOLOGY	1
AUTOMATION AND CONTROL SYSTEMS	1
PUBLIC ADMINISTRATION	1

3. Literature Review

This literature review will identify and present the current knowledge, theories, models, and discoveries in the field of research on the integration of AI in PM, through the examination of selected articles, which are categorized into project management areas of knowledge. The ten knowledge areas described in the PMBOK guide are Project Integration Management, Project Scope Management, Project Schedule Management, Project Cost Management, Project Quality Management, Project Resource Management, Project Communications Management, Project Risk Management, Project Procurement Management, and Project Stakeholder Management [24].

This classification aims to highlight the evolving role of intelligent systems and methodologies in enhancing decision-making, predictive accuracy, and overall project outcomes. In addition to identifying gaps, contradictions, or difficulties in current research, this chronological analysis will aid in understanding the field's evolution and contributing to identifying areas where further research is needed.

3.1. Project Integration Management

This literature review describes 13 articles that could be categorized under the area of Project Integration Management. More specifically, focusing on monitoring the progress of construction projects, the authors review pattern recognition techniques, which include neural network pattern recognition (NN-PR) techniques and statistical pattern recognition (S-PR). Integrating AI into pattern recognition has enabled project managers to access advanced features for decision-making within the construction industry [25]. In the same industry, a hybrid model was developed, the evolutionary fuzzy hybrid neural network (EFHNN), that merges four AI approaches: NNs, high-order NNs, FL, and GAs. This model is used in the execution phase of projects in order to assist project managers in controlling project performance and in implementing corrective measures necessary for project success [26]. From planning to execution and monitoring, a new proposed multivariate model suggested identifying performance issues and helping project managers to implement corrective measures [27]. Fuzzy time series forecasting and data envelopment analysis were used to create a theoretical framework to improve the prediction of project performances. Futures research has suggested the incorporation of ML algorithms in order to improve predictive capabilities [28]. For software projects, an entailment-based intelligent system was proposed (EIS), which contains NLP, entailment recognition, contradiction recognition, a recognition result merger, and a performance decision-maker and aims to monitor and manage projects in the monitoring and control phase of projects [29]. Focusing on the creation of a model capable of predicting the performance of construction projects under unit price contracts, another study suggested the use of the System Dynamics (SD) approach. Futures research has incorporated the integration of real-time data and sensor technologies to enhance the accuracy and timeliness of performance forecasting in construction projects [30]. Cheng et al. [31] created the Hybrid Gaussian Process Inference Model, which consists of data input, the Gaussian process and Bayesian inference, and hyper-parameter optimization. This model offers accurate forecasts regarding the results of construction projects to help in the decision-making process. FL and hybrid fuzzy approaches have been applied to address uncertainty in construction variables. Fuzzy set theory has been used to represent transitions between concepts, improving the realism of predictions. By combining fuzzy arithmetic with ML, this approach seeks to boost decision-making and project outcomes in construction engineering and management. Future studies could explore methods like conditional fuzzy clustering, collaborative fuzzy clustering, DL, and hierarchical learning to manage the complexity of construction problems caused by high dimensionality, to enhance the accuracy of fuzzy hybrid models [32]. A new framework

combined Building Information Modeling (BIM), ML algorithms, Virtual Reality (VR) technology, and the Unity game engine to create a game-like hybrid application for the on-demand monitoring of construction projects. By focusing on enhancing integration, real-time data processing, automation, scalability, user experience, interoperability, and validation, future research could improve PM and decision-making in the construction industry [33]. Aiming to automate the alignment of long-term and short-term plans in construction projects, a hybrid model has been proposed that consists of NLP techniques. By using this model, managers can improve the productivity, effectiveness, and precision of construction projects [34]. Moreover, a new model was created, which integrates GAs and probabilistic models, to forecast the outcomes of agile projects within global software development enterprises. According to the creators, this model could be used by project managers in the planning stage of a project to forecast its success [35]. The same year, Uddin et al. [36] proposed five ML models—logistic regression, the SVM, K-Nearest Neighbors, the Random Forest, and Extreme Gradient Boosting—to describe the connections between project attributes, networks, and the “Iron Triangle”, which contains project cost, time, and quality. Different models outperformed others, based on the inputs. These models could be used by project managers in different stages of projects. More specifically, by using the models in the planning stage, managers could predict duration and project cost; in the execution phase, they could control project progress and risks; and in the control phase, they could minimize risks and modify project plans. Kim et al. [37] developed an alternative method that can be used to measure and monitor the performance of construction projects in situations where credible data are collected for performance indicators such as cost duration using GAs.

3.2. Project Scope Management

Project Scope Management includes two articles. The first one focuses on suggesting a hybrid model called BABE, which combines the Artificial Bee Colony (ABC) and Analogy-Based Estimation (ABE) approaches. Focusing on software development projects, this model could be used in the early stages of a project in order to foresee the effort needed for development [38]. Another study explores the use of the Extreme Learning Machine (ELM) for estimating software development effort and concludes that it outperforms other methods in accuracy, based on metrics like the mean absolute error (MAE) and the magnitude of the relative error (MRE). Future research could focus on improving the model through parameter tuning, adding more machine learning techniques, and testing on larger, more varied datasets [39].

3.3. Project Schedule Management

In 2011, research was conducted in order to propose an ant colony optimization algorithm that could improve existing software quality assessment models by adapting them to new software systems. Ant colony optimization (ACO) is a swarm intelligence algorithm that takes inspiration from the foraging behavior observed in real ant colonies [40]. This model excels in comparison to other approaches, such as the C4.5 ML algorithm and random guessing, while maintaining the transparency of the models it generates [41]. The same year, in order to forecast the duration for the reconstruction project following the Wenchuan earthquake, two ML models were developed, using Bromilow’s time–cost (BTC) model and the Elman network (EN). Even though the EN model was more accurate in reconstruction time than the BTC model, the authors concluded that the BTC model is easier to use in real-life projects [42]. Another study used an ANN to predict project schedules in the Indian construction industry. The developed model exhibited strong predictive skills [43]. Irfan et al. [44] decided to develop mathematical models in the planning stage in

order to estimate the duration of highway projects by creating mathematical relationships between the duration and the factors that affect it, such as project cost, project type, and contract type. The same year, a neurogenetic approach was proposed, which combines the strengths of GAs and NNs, focusing on project scheduling. This hybrid approach improves the quality of solutions and contributes to the advancement of the field of project scheduling by enhancing the efficiency of scheduling processes [45]. Chou et al. [46] suggested a hybrid intelligence model, called ESIM, Evolutionary SVM Inference, which consists of an SVM and a fmGA. The model was designed for application in the initial stage of software development projects and surpassed traditional models in terms of estimating the time required to complete the development of a software project. Sadeghi et al. [47] developed a fuzzy discrete event simulation (FDES) framework for the construction industry. This framework combines FL with discrete event simulation to refine the precision of project time estimation. The accuracy of the proposed FDES model was attributed to the approach for calculating event times. For future research, the authors suggested the addition of algorithms and approaches in order to improve project scheduling and management. Wauters and Vanhoucke [48] recommended a hybrid model, which contains the Nearest Neighbor (NN) method, decision trees (DTs), Bagging, the Random Forest (RF), Boosting, and the SVM. During the planning phase of a project, this model aims to predict the duration of the project. Focusing on dam projects, Golizadeh et al. [49], in order to predict the duration of a construction project, created a multilayer perceptron network based on the backpropagation learning method using an ANN. This model was used in the pre-contracting phase of a project. For software projects, Choetkiertikul et al. [50] used a variety of ML approaches, such as Random Forests, NNs, decision trees, the Naïve Bayes method, DNNs with dropouts, and Gradient Boosting Machines, in order to build models that could forecast whether an issue was at risk of being delayed. In the year 2018, Pospieszny et al. [51] suggested the combined use of three ML algorithms, the SVM, the multilayer perceptron (MLP), and generalized linear models (GLMs). This research aimed to estimate software project duration and effort in the initial stages of a project. In addition, a hybrid model, called the firefly-tuned least squares SVM (FLSVM), which combines the least squares SVM (LS-SVM) and the firefly algorithm (FA), was suggested in order to forecast the duration of the construction of a diaphragm wall [52]. In the construction field, in order to predict delays in construction projects, the impact of factors causing delays can be measured and the time performance of a project can be forecasted according to risk levels. The model named the Random Forest classifier with genetic algorithm optimization (RF-GA) could be used in the planning and scheduling stages of a project [53]. Gondia et al. [54] developed and compared the performance of two models: one based on the decision tree algorithm and the other based on the Naïve Bayesian algorithm. The evaluation and comparison of these two models allowed the researchers to conclude that the Naïve Bayesian model provided superior predictive performance for analyzing project delays in the construction industry compared to the decision tree model. Research could focus on creating dynamic risk analysis models that could adjust to evolving project conditions and offer real-time risk assessments throughout a project lifecycle. In 2021, a new model was created in order to forecast delays in construction projects. This model used Random Forest classification and real-time data to assist in updating managers on a project's schedule in real time [55]. In the context of construction schedule management, a subset simulation-based Probabilistic Critical Path Method (PCPM) was introduced, in order to provide to project managers the ability to monitor timetables and reduce the risk of project delays [56]. The same year, a DL-based approach that combined long short-term memory (LSTM) and a gated recurrent unit (GRU) proposed to predict difficulties in the scheduling of engineering projects in order to help managers take corrective measures to guarantee the on-time

completion of them [57]. The scheduling risk assessment framework (SRAF) addresses uncertainties in project duration and resource availability by capturing, analyzing, and integrating probabilistic distributions related to activity durations and resource breakdown scenarios. The focus of this framework is on enhancing project schedules by incorporating buffer time, adjusting the objective function, and applying the Enhanced Move-Based Local Search Heuristic (EMBLSH) algorithm to manage uncertainties and enhance project results. Future studies could aim to find a way for how this model could help practitioners predict future budgets more accurately, reduce financial and time losses, and improve project-planning strategies [58]. By using eight ML algorithms to train regression models, researchers aimed to predict the effort and duration of tasks in software projects. After creating 48 different regression models, they concluded that the Random Forest, the Extra Trees Regressor, and XGBoost consistently outperformed the other algorithms [59]. In order to monitor and forecast a Graphical Evaluation and Review Technique (GERT) project, a hybrid model was proposed, which combined stochastic EDM (Earned Duration Management) and ML algorithms, including Ada Boost, the Bagging Regressor, and Gradient Boosting. The model used the Monte Carlo simulation to collect information for the project completion time [60]. Short-term project progress could be predicted by a comprehensive framework that combined data analytics techniques with agility practices, as explored by Jafaar et al. [61]. Moreover, another model developed in a study used fuzzy PERT (FPERT) and the critical path method (CPM) to provide accurate timeline and budget estimates for digitized startups, specifically focusing on the case study of the startup e-Karsaz in Pakistan. Future research in this area could focus on validating and refining the proposed model by conducting comparative analyses with other startups and integrating additional PM techniques to enhance accuracy and applicability across various startup contexts [62]. Yang and Chen [63] focused on elevating Earned Schedule Methods (ESMs) in order to forecast project duration for delayed projects by proposing the ESmin+ method. Future studies should focus on validating this method, using actual projects to assess its effectiveness in more complex and real-world scenarios. These articles focused on enhancing Project Schedule Management.

3.4. Project Cost Management

Many authors have tried to explore the area of Project Cost Management. Cheng and Roy [64] created a hybrid Support Vector Machine (SVM) model, EFSIM T (Evolutionary Fuzzy SVM Inference Model for Time Series Data), to predict cash flow and control project performance. This model was used in the planning and implementation phases of a project with better outcomes than other models and consisted of three AI methodologies, FL, weighted SVMs, and fast messy GAs (fmGA). Hwang [65] developed the time series models ARMA (5,5) and VAR(12) in order to forecast construction costs. Utilizing time series index data for cost estimation enhances forecasting accuracy, supports informed decision-making, and enables proactive cost management strategies based on historical cost trends and market dynamics. Moreover, another study suggested the use of a novel time-dependent evolutionary fuzzy SVM inference model that combined FL, a weighted SVM (wSVM), and a fmGA. By combining these methodologies, the authors aimed to enhance the performance of Estimation at Completion (EAC) in construction projects. The use of additional ML methodologies could enhance the predictive capacity of the model and the use of real-time data could provide more precise data to improve cost estimation [66]. In the computer science community, a study was published, which presented a hybrid model, a long-linear regression and multilayer perceptron (MLP) NN model. In the early stages of software projects, this model aims to offer early cost estimation [67]. In the year 2014, a new research work forecasted the accuracy of projects' Earned Value metrics by

developing a mathematical modeling procedure and combining time series and linear regression analysis. The authors suggested the combination of AI, ML, or big data analytics in order to enhance the accuracy of the results [68]. A year after, the Second Moment Bayesian (SMB) model was created, which could be applied in the execution phase of a project in order to predict the final project cost, using data collected during the project's execution [69]. The authors tried to create a model for Taiwanese construction projects. More specifically, a hybrid computational model for forecasting the Taiwanese Construction Cost Index (CCI) was proposed, which consisted of multivariate adaptive regression splines (MARS), a radial basis function neural network (RBFNN), an Artificial Bee Colony (ABC) algorithm, a generalized likelihood ratio (GLR) test, and an Extreme Learning Support Vector Machine (EL SVM). The hybrid model was created in order to assist cost engineers in dealing with the variability of the CCI and improving the accuracy of the CCI forecast model [70]. Another study compared four different NN models for software development effort estimation: the multilayer perceptron (MLP), general regression neural network (GRNN), radial basis function neural network (RBFNN), and cascade correlation neural network (CCNN). The findings indicated that the cascade correlation neural network (CCNN) and radial basis function neural network (RBFNN) tended to enhance the accuracy of software development effort estimation in comparison to the multilayer perceptron (MLP) and general regression neural network (GRNN) models. Further investigation involving alternative validation methods such as leave-one-out (LOO) validation could offer more information on the performance of these models [71]. In order to enhance cost forecast accuracy, another study suggested the combined use of Earned Value Management (EVM) models, Reference Class Forecasting (RCF), and the Exponential Smoothing Method (ESM). The study introduced the Exponential Smoothing with Reference Class Forecasting (XSM) method [72]. For the construction field, and specifically for expressway projects, a convolutional neural network (CNN) model was created, as a tool for the estimation of projects' cost. Before the determination of funding, a manager could apply this model in order to minimize capital risks and make considered decisions [73]. In another study focused on predicting project cost overrun levels, an ensemble-learning model employed the Ripple-Down Rule Learner (RIDOR) and K-star algorithms as foundational learners. The RIDOR is capable of automatically generating classification rules from input data, whereas the K-star model is a lazy classifier that retains training instances for classification purposes. These algorithms were chosen based on their proven reliability and accuracy in previous construction data studies, thus enhancing the effectiveness of the ensemble model in predicting cost overrun levels. The incorporation of real-time data, such as project progress updates, market trends, and economic indicators, into the predictive models could enhance their accuracy and responsiveness to changing conditions [74]. In the field of construction, a hybrid model, consisting of multiple ANN models that were trained by the Lower Upper-Bound Estimation (LUBE) method, was created to forecast material prices [75]. Xue et al. [76] discussed integrating the Sparrow Search Algorithm (SSA) with a backpropagation neural network (BPNN) to enhance cost prediction accuracy for substation projects. By optimizing the BPNN's weights and thresholds, the SSA-BP model outperformed traditional approaches, achieving lower error metrics. Future research should focus on incorporating qualitative factors, improving data collection, and broadening the model's application to better analyze and predict costs in the energy sector. Kim and Cha [77] introduced a hybrid expert system known as LD-MCS, which integrates a rule-based system, logic-based design (LD) with a case-based reasoning system, and a memory-based case system (MCS) for the automated prediction of costs in apartment renovation projects. This system is engineered to enhance the precision and efficiency of cost estimation through the fusion of expert insights and historical data. Focusing on cost

estimation, Castro Miranda et al. [78] published a systematic literature review in order to analyze predictive analytics methods that can improve the effectiveness of estimating construction costs during the initial phases of projects. Lee et al. [79] proposed an advanced ANN-based cost estimation model that could integrate ensemble modeling and factor analysis to enhance the precision and robustness of preliminary cost estimates for large-scale construction projects with limited historical data availability. Futures research has suggested the use of additional ML techniques and algorithms for cost prediction models with limited data samples. The evaluation of ML-based techniques in calculating the effort and duration of individual tasks or issues in software projects was studied in another study. Regression analysis was conducted and a parametric model was used in order to estimate the cost of new product development projects [80]. A combination of the fuzzy analytic hierarchy process (FAHP) and traditional ML techniques, such as K-Nearest Neighbors (KNNs), Random Forests (RFs), and XGBoost, was proposed to create a Construction Cost Index (CCI) specific to Jordan's construction industry and predict upcoming CCI values. Future research could focus on new ML algorithms or hybrid models for improved accuracy and in predicting the integration of the construction costs of advanced technologies such as AI, the Internet of Things (IoT), and big data analytics [81].

3.5. Project Quality Management

Project Quality Management was analyzed by Chen et al. [82], who created a decision tree framework that was designed to enhance the evaluation and quality assurance of software processes. In 2019, a hybrid model was developed that consisted of FL and NNs, the weighted neuro-fuzzy framework. This model is applied in the early and later stages of projects in order to forecast errors in the software and to enhance the quality of projects by using different data [83]. A Bayesian Belief Network (BBN) model was used to forecast project performance, based on historical data, as it was more accurate in identifying complexity, uncertainty, and performance factors [84]. With the use of data mining, researchers aimed to detect elements that participate in project success. Using these data, an ANN was proposed that could predict the success of projects based on these factors [85].

3.6. Project Resource Management

Project Resource Management was explored in 2011, when ANNs, Gas, and FL were combined to create the neuro-fuzzy genetic system and improve the accuracy of selecting competent project managers for construction projects [86]. A multilayer feedforward NN trained with a backpropagation algorithm was used in the implementation, planning, and design stages of a project to measure and forecast workforce efficiency in construction projects [87]. Patel and Jha [88] proposed an ANN model that aimed to predict the safe work behavior of employees in construction projects. In another study, a hybrid model based on the ANN and activation and transfer functions was created in order to forecast, improve, and manage labor productivity in construction projects. The model was trained to use everyday data from the employees' work hours [89]. In addition, an NLP was applied in order to create a text classification model that could automatically allocate building service requests to maintenance personnel. The NLP was combined with three ML methodologies to find the most accurate model. Between logistic regression, the Naïve Bayes method, and the SVM, the last model surpassed the other two [90]. In the year 2022, a hybrid neuro-fuzzy system that was optimized using an innovative EEBAT algorithm was applied in projects that used scrum agile PM techniques to help managers in selecting suitable resources. This was achieved by minimizing the difference between actual and estimated effort while ensuring timely delivery within defined timelines [91]. FL and

metaheuristic approaches have been combined to investigate the strength, energy efficiency, and cost-effectiveness of building materials in construction management. ML algorithms with existing hybrid metaheuristic approaches could develop the accuracy and efficiency of material optimization in construction projects in future studies, while developing a real-time monitoring and control system would help the algorithm to monitor building material usage based on changing environmental conditions and energy demands [92]. A model was presented that offered a mathematical framework, which addressed the intricate interplay among construction tasks, resource allocation, financial flows, and the uncertainties arising from various influencing factors. The model was based on a partially observable Markov decision process. Furthermore, the model harnessed deep reinforcement learning (DRL) to achieve the ongoing adaptive optimization of labor and material management, leading to the enhancement of work and cash flows. Additionally, a discrete event simulation-based simulator was created to replicate dynamic project characteristics and external conditions, contributing to the training of deep reinforcement learning [93]. Another study used a total of seven ML models to predict organizational agility. These models included the SVM, decision trees (DTs), K-Nearest Neighbors (KNNs), the Gradient Boosting Machine (GBM), the Random Forest (RF), the Naïve Bayes (NB) method, and logistic regression (LR). The results of the study indicated high test accuracy percentages for most of the models, with the GBM, RF, SVM, and KNNs ranking higher than the other methods [94].

3.7. Project Risk Management

The ANN was used to examine the decision process regarding risk in public–private partnership (PPP) projects. Even though the model produced satisfactory results, the authors suggested, as future research, the use of other approaches to create more accurate models [95]. In the year 2012, another hybrid model was created, which consisted of a casual map and a Bayesian Network (BN), in order to forecast how combinations of alterations in variables of interest (leadership style, agility, project performance) influence overall project performance [96]. In the same year, a study was published that focused on early planning. More specifically, a hybrid model was suggested that combined NNs and SVMs with traditional logistic regression models. Project teams could use these models in the early planning stage of a project to better predict final results [97]. Fang and Marle [98] focused on developing a simulation-based risk network model within a decision support system. They created a framework in order to offer to project managers the ability to identify, analyze, model, and respond to risks effectively. The authors suggested the use of real-time data to develop a real-time risk monitoring system for future research. In the year 2013, a framework for software project risk analysis and planning was proposed that combined risk analysis and planning in the early stages of a project. The framework used ML algorithms (Random Forest, NN, and logistic regression) to achieve risk analysis and estimations [99]. Shi et al. [100] suggested FL theory, as a method, in order to analyze and manage delivery risks in construction programs. Futures research combines the integration of advanced technologies to improve decision-making in program management. Focusing on FL and risk management, like the previous study, another study aimed to enhance project performance by optimizing risk allocations [101]. Gunduz et al. [102] proposed a model, which contained FL, as a decision support tool for contractors to identify delays in Turkish construction projects before the bidding stage. In 2016, a new method was proposed, which combined the Analytic Hierarchy Process (AHP) and fuzzy inference system (FIS) in order to offer a more adaptable method for risk evaluation in IT projects [103]. A hybrid model called the BN-HFACS combines the Human Factors Analysis and Classification System (HFACS) with Bayesian Network (BN) modeling to forecast safety performance in construction projects [104]. Targeting risk management,

FL combined with mathematical models aimed to assess the probability of risk events and their consequences on the financial stability of a company, particularly in the context of promoting software products [105]. Focusing on Build–Operate–Transfer (BOT) projects, authors proposed a fuzzy analytic hierarchy process (FAHP) model that joined fuzzy theory with the analytic hierarchy process (AHP) to determine the importance of stakeholders' risks [106]. A combined approach using Social Network Analysis (SNA) and the Matrice d'Impacts Croisés Multiplication Appliquée à un Classement (MICMAC) method was employed to investigate and comprehend factors influencing project delays. Futures research has involved AI methods like DL and reinforcement learning to improve the predictive abilities of models used for analyzing project delays [107]. Assaad et al. [108] developed a framework that focused on predicting project performance based on project risks. The model contained mathematical and statistical analysis techniques. Further research is needed in order to determine the importance of project risks in cost and schedule overruns. Practical application and validation with real-world data are required in order to validate the model and compare it with existing predictive models. In the same scientific area of construction, focusing on forecasting financial losses due to natural processes, a deep neural network (DNN) was developed, using the convolutional neural network approach [109]. ML techniques were used in order to create models that could predict conflicts in construction projects. Among all the models, the SVM was the most accurate. Project managers could benefit from this model by identifying risks and factors that cause dispute appearances and proceeding to measures that minimize the problems [110]. A hybrid approach combining machine learning techniques, specifically a genetic algorithm (GA) integrated with K-means clustering, focused on identifying high-risk factors in large-scale construction projects. By analyzing expert-sourced data and employing the synthetic minority over-sampling technique (SMOTE) for better data representation, the study effectively categorized risks based on their severity and impact on project performance, enhancing stakeholder awareness and risk allocation [111]. For the purpose of forecasting cost overruns in thermal power plant projects, focusing on the most critical risk factors, a hybrid model was created that combined genetic programming (GP) and Monte Carlo simulation (MCS). By using real data from projects, this model could be applied in the early stages of a project to help managers confront risk efficiently [112]. Moreover, another group of researchers proposed a hybrid model, a combination of wavelet analysis and an artificial neural network (W-ANN), to forecast the incidence of accidents in the construction industry [113]. Researchers utilized the DNN algorithm to develop a model, which could predict financial damages from accidents in apartment construction sites. The model was trained by historical data and the predictions differed based on the inputs that the user included [114]. In addition, Santos et al. [115], focusing on assisting project managers in understanding the relationships between project activities and outcomes, in order to distribute resources and manage risks, created a new method that involved ML techniques, specifically Monte Carlo sampling and the triad method, alongside the Shapley value approach from cooperative game theory. Finally, another study expanded the development of a construction site dynamic risk ensemble model, based on ML algorithms, that could offer forecasts safety risk indicators in various workplace settings. The ensemble model, which consisted of the Naïve Bayes model, decision trees, Random Forests, SVMs, and ANNs, combined the prediction capabilities of the individual algorithms to make predictions, which could achieve the same level of performance as the top-performing base algorithms or potentially outperform them [116]. De Andrade et al. [117] provided project managers with a framework for assessing and enhancing project forecasting accuracy by leveraging the concept of project consistency. Advanced data analytics techniques, such as ML algorithms, could be added in the future to improve the prediction accuracy of project

duration forecasting based on project regularity indicators. All the articles above could be categorized under the area of Project Risk Management.

3.8. Project Stakeholder Management

Project Stakeholder Management was analyzed by focusing on improving decision-making and conflict management in construction projects. The study proposed a framework that integrates SVMs, FL, and the fast and messy genetic algorithm (fmGA). This hybrid AI system outperforms the use of Support Vector Machines (SVMs) [118]. In the same industry, a study suggested a multilayer perceptron (MLP) NN model, which could improve the decision-making process for arbitration, litigation, and contract administration by concerning differences in contracts [119]. Emphasizing the use of data and ML, a data-driven framework was suggested to assist project managers in decision-making and problem solving. This framework could adapt to different projects, and its accuracy and effectiveness depend on its application in specific problems and datasets [4].

3.9. Project Cost and Schedule Management

Project Cost and Schedule Management, combined in the construction industry, were studied by utilizing a framework of Earned Value Management. Future studies could focus on adding ML algorithms and AI methodologies to enhance the accuracy of project duration and cost predictions [120]. In addition, an SVM regression model was created to control project time and cost outcome forecasting. This framework helped project managers make decisions about planning, execution, and monitoring [121]. Habibi et al. [122] proposed a new hybrid model that they aimed to apply in the planning and control stages of projects in order to improve time and cost estimation and project scheduling. The model combined classical PERT and CPM methods with FL and appeared to offer more accurate estimations than traditional methods. A model described in another study suggested the use of actual project schedule information to assess how well Earned Value Management (EVM) and Earned Duration Management (EDM) could predict project outcomes. The study analyzed the effectiveness of project duration forecasting techniques [123].

3.10. Evolution Path

By examining the literature, Figure 3 explores how AI's contributions to PM have developed over time, highlighting key milestones and emerging trends in research. More specifically, between 2011 and 2012, research focused on introducing basic AI techniques like ACO and hybrid neural networks for scheduling and cost estimation. From the year 2013 to 2016, studies expanded into hybrid systems by analyzing the growth of advanced algorithms combining fuzzy logic, neural networks, and genetic algorithms for risk and resource management. Later, in 2017–2019, researchers marked a shift towards integrating AI with real-time data and multivariate analysis for execution monitoring and performance improvement. The integration of NLP, the IoT, and DL for enhanced communication, prediction, and quality management was expanded between the years 2020 and 2021, while current trends (2022–present) focus on holistic frameworks that integrate multiple PM knowledge areas while addressing challenges in stakeholder and integration management.

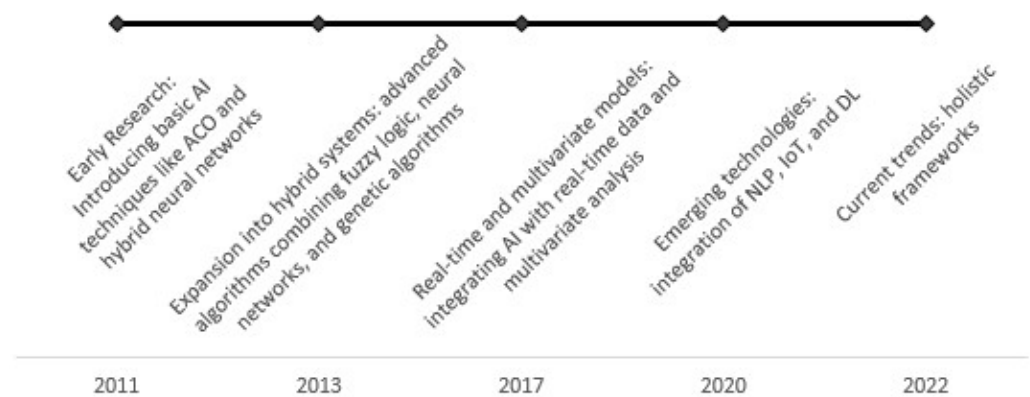


Figure 3. Evolution path of AI techniques in PM.

3.11. Decision Tree Analyses

Focusing on the literature above, this study introduces a set of decision trees that were developed to offer a guide for comprehending the ways in which different AI methodologies are used during the planning, execution, and monitoring stages of a project. They provide a summary of the relationships among several AI models, including machine learning, deep learning, and hybrid models, as well as the applications of them in projects.

The first decision tree, shown in Figure 4, aims to illustrate how several hybrid AI models combine strategies to enhance PM outcomes, such as cost, duration, and risk. A “hybrid model” in machine learning is a method that combines multiple algorithms or techniques to improve performance by leveraging the strengths of each component. These models are valuable in fields requiring high accuracy, like finance and engineering, where they enhance predictive power and data handling [124]. The tree in Figure 4 highlights the application of AI and hybrid methods in PM, targeting enhanced efficiency in project duration, cost, and risk management. For duration optimization, models combine neural networks, genetic algorithms, fuzzy logic, and machine learning, aiming to predict or optimize scheduling and forecasting. For cost optimization, combinations include Bayesian techniques, fuzzy logic, and regression models to enhance cost prediction accuracy. Risk analysis methods incorporate simulation (Monte Carlo) and AI for robust risk assessment, with predictive models such as decision trees and logistic regression applied to identify and classify risk factors. Figure 4 expands in detail in Figures 5–7. In Figure 8, AI models are categorized according to the project phase they are most effectively used in (planning, execution, monitoring). Through this structure, readers can understand the context and timing of applying different models to achieve the best results, along with an organized plan for effectively applying AI across the project lifetime.

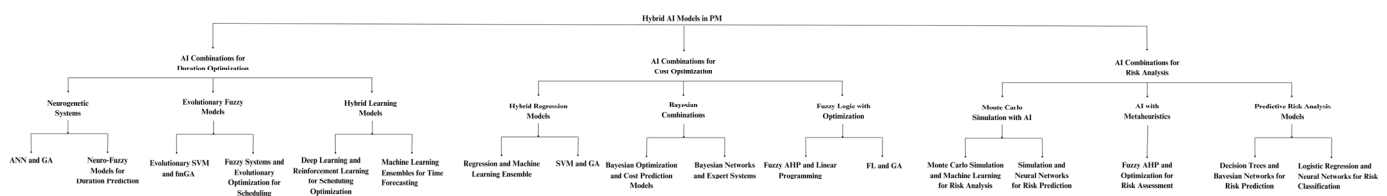


Figure 4. Different hybrid AI models combine techniques for PM outcomes.

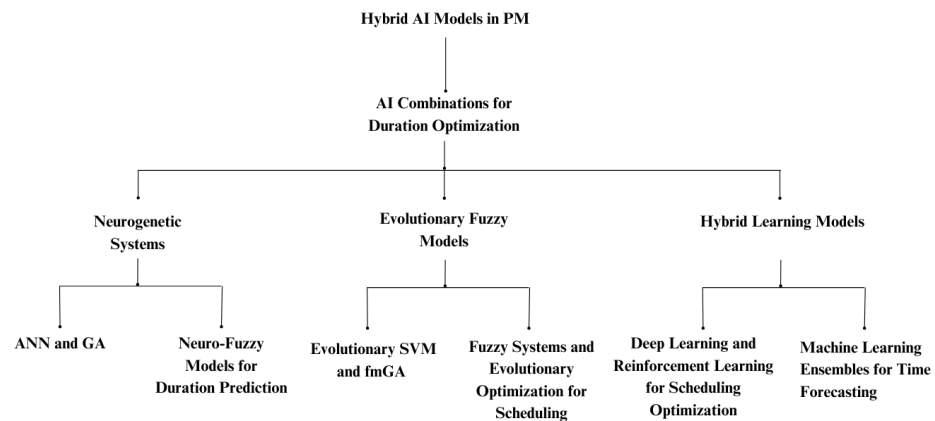


Figure 5. Part of Figure 3—hybrid AI combinations for duration optimization.

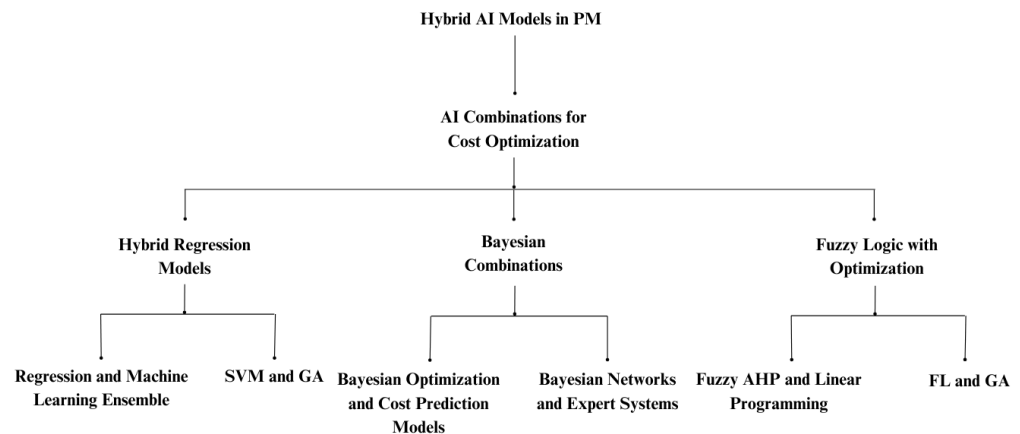


Figure 6. Part of Figure 3—hybrid AI combinations for cost optimization.

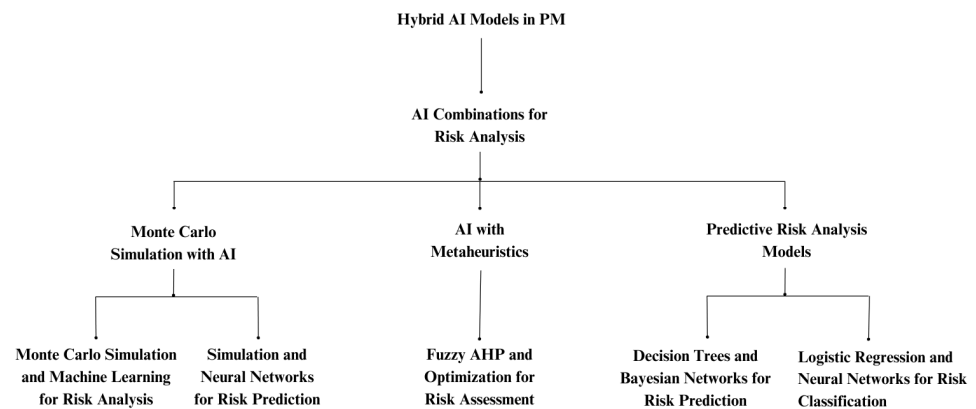


Figure 7. Part of Figure 3—hybrid AI combinations for risk analysis.

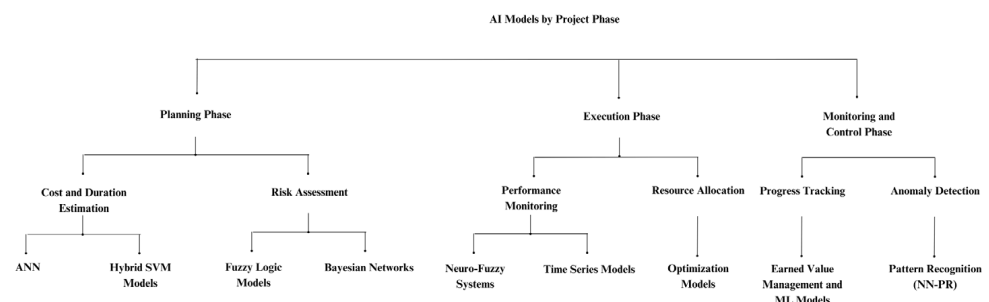


Figure 8. AI models according to the project phase they are most effectively used in.

Figures 9–11 focus on cost, duration, and risk forecasting, respectively. Figure 9 illustrates the various AI methodologies used for cost estimation in PM. This tree provides project managers with a comprehensive understanding of which methodologies are most appropriate for their particular requirements and project contexts by classifying models into categories such as regression, neural networks, Bayesian methods, and expert/statistical models. Figure 10's decision tree classifies the AI models used to predict project durations. It makes a distinction between hybrid, deep learning, and more conventional machine learning techniques, such as regression and time series models. Each branch focuses on particular methods and how they might be applied. The AI models used in Project Risk Management and forecasting are highlighted in Figure 11. It illustrates how several AI methodologies are applied to identify, assess, and minimize risks. From FL systems for risk prediction to complex neural networks for performance forecasting, the decision tree shows how techniques for risk management evolve as a result of the application of AI methods.

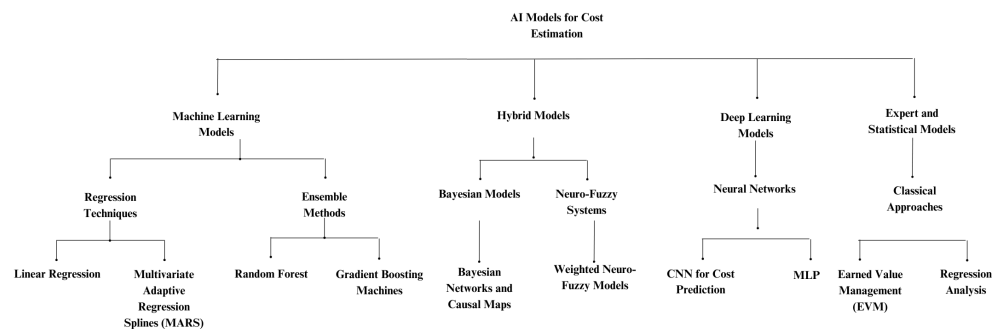


Figure 9. AI methodologies used for cost estimation in PM.

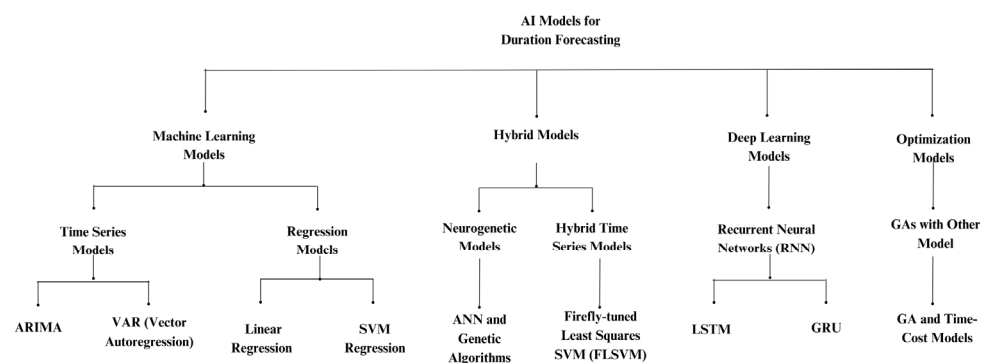


Figure 10. AI models used for forecasting project durations.

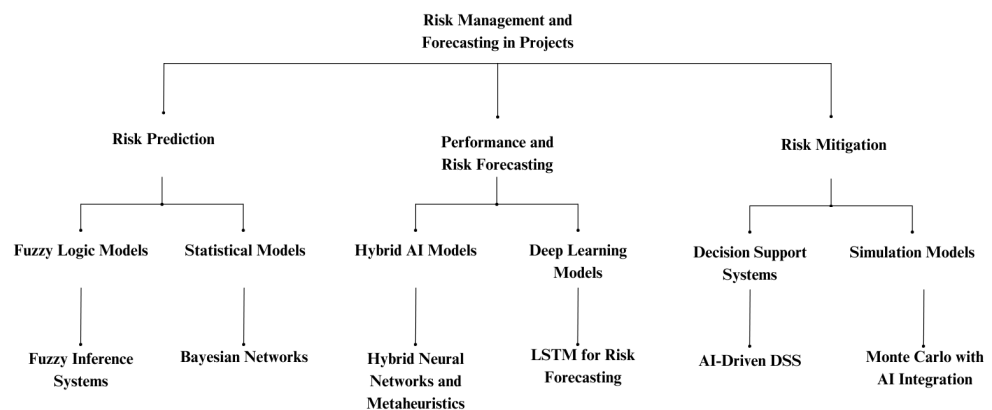


Figure 11. AI models used for risk management and forecasting within projects.

3.12. Data Quality and Sources in AI-Driven Project Management Studies

The success of AI methodologies is heavily contingent upon the quality of the data utilized, highlighting challenges such as data's completeness, accuracy, representativeness, and temporal relevance [125]. The studies in this systematic review have used a range of data sources, such as industry reports, structured datasets, and case studies of real-world projects. Many studies rely on publicly accessible databases in order to ensure transparency and consistency in their findings. However, when it comes to actual project restrictions, these statistics frequently lack detail. Although they provide high-quality, accurate information, industry datasets are sometimes inaccessible because of confidentiality agreements. Certain studies have chosen to use predictive models to simulate data, which enables controlled testing but might not accurately represent the unpredictability of the real world. Additionally, expert interviews and survey-based data have been used to obtain qualitative insights into project management, although subjectivity and bias may have been included. Several studies have highlighted issues with completeness, where missing data points affect the reliability of AI-driven predictions, particularly in cost estimation and risk assessment models. Accuracy remains a significant concern, as project parameters such as budget, timeline, and risk factors often fail to align with actual project outcomes. While some studies incorporate real-time IoT-driven data collection, many still rely on static historical datasets. This can create discrepancies in how AI-driven predictions align with evolving project environments. Studies leveraging a hybrid approach, combining structured data with real-time and expert-driven insights, tend to provide the most comprehensive datasets. Focusing on the articles contained in this literature review, it is highlighted that the cross-industry validation and benchmarking of AI models remain largely underutilized. This undermines the real-world applicability of the suggested AI-driven project management tools.

4. Conclusions, Discussion, and Future Work

This literature review presents the development of PM methodologies by examining the use of hybrid models, such as ML and FL, to improve project outcomes and predictive accuracy. Research has indicated that models such as the weighted neuro-fuzzy framework and the BABE model can be helpful for project effort and software error predictions. Furthermore, in the software development and construction industries, advanced methodologies like deep learning and Bayesian Networks have proved crucial for risk assessment and performance forecasts. Models such as the Random Forest and XGBoost consistently outperform other models, highlighting the trend toward real-time data integration for accurate predictions.

AI's contributions to project management have been substantial, particularly in Project Schedule Management, Project Cost Management, and Project Risk Management. Predictive models such as machine learning algorithms and hybrid systems have participated in improving forecasting accuracy, optimizing schedules, and identifying risks. By classifying the literature into the 10 areas of knowledge in PM, it is concluded that limited research has been conducted on Project Integration Management, Project Scope Management, Project Quality Management, Project Communications Management, Project Procurement Management, and Project Stakeholder Management.

Despite notable improvements in hybrid models and ML applications for PM, several gaps and weaknesses become visible in the literature. A large number of studies are not fully validated using real-world data, which reduces their practical relevance. Furthermore, while AI is increasingly used in planning, scheduling, and execution, there is little focus on post-project performance. Additionally, many current models neglect the interdependencies between different risk indicators in favor of a more fragmented approach to risk assessment.

Knowledge areas like Project Communications Management and Project Stakeholder Management require skills, emotional intelligence, and contextual awareness, qualities that are challenging for AI systems to replicate. On the other hand, research has focused on quantitative aspects of project management, such as cost estimation and schedule management, as these align more effectively with the capabilities of existing AI models. Implementing AI in underexplored areas often demands significant investments in infrastructure, skilled personnel, and organizational change, which can create barriers for both researchers and practitioners. Additionally, resistance to adopting AI in areas involving human interaction and decision-making has further slowed progress. Concerns over job displacement and the interpretability of AI-driven decisions exacerbate this resistance, contributing to the persistent gaps in AI's application across all project management knowledge areas.

Several limitations remain in the application of AI to project management. One challenge is the overfitting of machine learning models, particularly when models are trained on historical project data that may not reflect the complexities and uncertainties of real-world projects. Another significant challenge is the generalization of AI-driven project management methodologies across industries. While AI applications in construction and IT project management have been well-explored, their applicability in other industries—such as healthcare, finance, or public administration—remains underdeveloped. Differences in requirements, project structures, and stakeholder interactions create barriers to widespread AI adoption. Moreover, the integration of AI into project management requires substantial investment in data infrastructure and expertise, which may be a barrier for smaller organizations.

Future research should expand on unexplored knowledge areas in PM by developing AI tools focused on Project Communications Management to analyze communication patterns, on Project Procurement Management to analyze contracts, and on Project Stakeholder Management to analyze preferences and concerns, while industry-specific AI adaptations and cross-domain validations should be formed to enhance AI's reliability in diverse project environments. AI-based structured frameworks could be developed for analyzing project success factors and feedback loops to enable continuous improvement. No previous research has investigated the application of a framework capable of integrating real-world data across multiple PM knowledge areas, such as risk, cost, and schedule management, to provide a new approach to project decision-making. Questions are raised for further research. The first question focuses on enhancing project forecasting and decision-making in PM by analyzing and incorporating real-world data into current AI models. The second question addresses the difficulties in maintaining the accuracy and consistency of AI models when constantly updating them with real-world data. The third question concerns how to improve decision-making and project performance in PM using predictive models based on AI. These models could simultaneously focus on developing the prediction and optimization of cost, duration, risk, and task prioritization.

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